Topology Exam 1 Cheatsheet

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Topological spaces

topology A set T of subsets of X such that

1. $X \in T$
2. $\varnothing \in T$
3. T is closed under *arbitrary unions* and *finite intersections*

basis A collection $\mathcal{B}$ of subsets of X such that

1. $\forall x \in X, \exists B \in \mathcal{B} : x \in B$
2. $\forall B_1, B_2 \in \mathcal{B}, x \in B_1 \cap B_2 \implies \exists B_3 \in \mathcal{B} : x \in B_3 \subseteq B_1 \cap B_2$

The set $T_{\mathcal{B}} = \{\bigcup \mathcal{F} \mid \mathcal{F} \subset \mathcal{B}\}$ is the topology generated by $\mathcal{B}$.

fineness T is finer than $S \iff S \subset T$

subspace (Y, T_Y) where $Y \subset X$ and $T_Y = \{Y \cap U \mid U \in T\}$

closure $\overline{A} = \mathrm{Cl}(A) = \bigcap \left\{U \mid \complement_X U \in T \wedge A \subset U\right\}$ is the smallest closed set containing A

interior $\mathrm{Int}(A) = \bigcup \{U \mid U \in T \wedge U \subset A\}$ is the largest open set contained in A

Special cases

discrete $T = \mathcal{P}(X)$

$\mathbf{T_2}$ $\iff \forall x_1, x_2 \in X, \exists$ neighborhood U_1 of $x_1 \exists$ neighborhood U_2 of $x_2 : U_1 \cap U_2 = \varnothing$

$\iff$ limits of sequences are unique

$\iff \forall$

$\iff$ attack

metric for metric space (X, d), the toTODOpology generated by $\{B_r(p) \mid r \in \mathbb{R} \wedge p \in \mathbb{R}^n\}$

order for $(X, <)$, the topology generated by the collection $\mathcal{B}$ of the following subsets $\forall a, b \in X$

- (a, b)
- $[\min(X), b)$
- $(a, \max(X)]$

On $\mathbb{R}$:

standard on $\mathbb{R}^n$, generated by $\{B_r(p) \mid r > 0 \wedge p \in \mathbb{R}^n\}$

upper

k-topology

Maps

continuous a map $f : X \to Y$ between topological spaces (X, T) and (Y, S) if $\forall V \in S, f^{-1}(V) \in T$

homeomorphism a bijection $f : X \to Y$ if both f and f^{-1} are continuous

Metric spaces

metric on a set X, the function $d : X \times X \to \mathbb{R}$ such that, $\forall x, y, z \in X$,

1. $d(x, y) \geq 0$
2. $d(x, y) = 0 \iff x = y$
3. $d(x, y) = d(y, x)$
4. $d(x, z) \leq d(x, y) + d(y, z)$

A metric *induces* the topology generated by the basis of all open balls using d.

metrizable a topological space if there is a metric that induces its topology

convergence